Laraib Mustafa Mirza

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Profile

M.Sc. graduate in Astrophysics and Cosmology at the University of Padua (Italy). In my thesis I modelled the radio-to-γ-ray emission of a $z = 3.65$ blazar to study its jet (emission processes, location of the emission region and jet power), and used the results to constrain the growth of its supermassive black hole. Hands-on experience: *NuSTAR* and *Fermi*-LAT data analysis, broadband SED construction, relativistic jet modelling (JetSeT) and Bayesian inference (MCMC).

Research Interests

High-energy astrophysics, X-ray and γ-ray astronomy, active galactic nuclei and supermassive black hole growth, relativistic jets, radiative processes, multi-messenger astronomy, and statistical methods for high-energy data.

Education

M.Sc. in Astrophysics and Cosmology 2026
University of Padua, Italy

B.Sc. in Physics 2021
Jamia Millia Islamia, New Delhi, India

Research Experience

Master's Thesis: Jet Physics of MG3 J163554+3629 and Its Impact on SMBH Evolution in the Early Universe
University of Padua, 2025–2026 — Supervisor: Prof. Eugenio Bottacini — Co-supervisor: Dr. Andrea Tramacere (Univ. of Geneva)

- Reduced and analysed *NuSTAR* hard X-ray data (32.8 ks) with HEASoft/NUSTARDAS; diagnosed background contamination in one module and fitted spectra in XSPEC (absorbed power law, $\Gamma = 0.94 \pm 0.34$; a broken power law is not statistically required).
- Assembled a radio-to-γ-ray SED (GBT/VLA, WISE, *Gaia*, *Swift*-UVOT, *XMM-Newton*, *NuSTAR*, *Fermi*-LAT) with systematics, upper limits and EBL absorption.
- Fitted a one-zone leptonic jet model with external disk, BLR and torus photon fields in JetSeT, with Minuit and Bayesian MCMC (emcee), over a grid of 7 accretion efficiencies; best fit $\chi^2 = 26.3$ for 27 d.o.f.; showed that the emission region lies outside the BLR.
- Showed that *NuSTAR* breaks the SSC/EC degeneracy: SSC provides 60–74% of the jet emission at 10 keV, and EC produces the spectral hardening towards 40 keV. Derived $M_{\rm BH} \simeq 1.8 \times 10^9\, M_\odot$, Eddington ratio $\lambda \simeq 0.08$, and a particle-dominated jet more powerful than the accretion disk.
- Combined jet energetics with black-hole growth models: the present accretion rate cannot explain the mass, which suggests an earlier phase of faster accretion at low radiative efficiency ($\eta 0.1$).

Code: github.com/Mirza-751/MG3-J163554-3629-SED-Modelling

Angulus: Spectrum-Centric GNSS Integrity Monitoring Nov 2025–present
Work Package Lead, Simulation & Validation — University of Padua (ref. Prof. Nicola Laurenti)

- Simulation and validation pipelines for GNSS spoofing detection.
- Selected for the ESA BIC Padua incubation programme (2026).

Academic Projects

- **X-ray timing and spectral analysis of a blazar** (supervised by Prof. Stefano Ciroi): 7 epochs of *Swift* XRT/UVOT data (2005–2022) of 1ES 2344+514; aperture photometry and XSPEC spectral modelling; identified flaring and long-term variability. github.com/Mirza-751/blazar-1ES2344
- **Supernova natal kicks of compact objects** (supervised by Dr. Giuliano Iorio): simulated 10^4 massive stars in a Milky Way potential with fireworks + SEVN; compared the Hobbs kick model at solar and sub-solar metallicity; neutron stars end up more widely distributed than black holes because of their stronger natal kicks. github.com/Mirza-751/nbody-neutron-star-black-hole
- **Cosmological perturbation theory** (supervised by Prof. Sabino Matarrese): derived the perturbed Einstein equations in synchronous and conformal Newtonian gauges (Ma & Bertschinger); built the gauge-invariant Bardeen potentials and connected them to CMB and matter power spectrum observables. github.com/Mirza-751/cosmological-perturbation-theory
- **Exoplanet transit detection** (supervised by Dr. Luca Malavolta): reduced 1857 frames from the Asiago 1.82 m telescope (differential photometry, 0.0038 scatter); TESS analysis with Wotan + TLS; MCMC with BATMAN refining $P = 1.4391$ d, $R_p/R_s = 0.1270$, $i = 84.09°$ for Qatar-1b. github.com/Mirza-751/qatar1b-transit-analysis

Technical Skills

High-energy data analysis: HEASoft, NUSTARDAS, CALDB, XSPEC (model comparison, F-test, χ^2 profiles, confidence contours); *NuSTAR*, *XMM-Newton* and *Swift* XRT/UVOT data; *Fermi*-LAT (Fermitools, fermipy); DS9; aperture and differential photometry.

Statistics & modelling: Bayesian MCMC (emcee), frequentist fitting (Minuit/iminuit), convergence diagnostics, posterior and degeneracy analysis; JetSeT (relativistic jet radiation modelling).

Programming: Python (NumPy, SciPy, Astropy, Matplotlib); C++, Fortran, C; Linux shell scripting; Git, Jupyter, LaTeX.

Languages: English (fluent, IELTS Academic 8.0), Hindi (native), Urdu (native), Italian (B1).

Schools, Workshops & Conferences

Early Universe from Home — virtual conference Feb 2026

X-Ray Spectroscopy Workshop — MIT, USA (virtual) Aug 2023

MW-Gaia School — Vilnius University, Lithuania Jul 2023

Talk: "Stellar Dynamics and Supernova Kick Simulations in the Milky Way"

JWST Training School — Instituto de Astrofísica de Canarias, Spain Jun 2023

Gravitational Wave Data Workshop — University of Padua, Italy May 2023

Other Experience

- **AI training (remote):** evaluation and alignment of AI reasoning on scientific tasks, including astrophysical plot interpretation — Alignerr (Jan 2026–present), Mindrift (Dec 2025–May 2026).
- Member, COST Actions CA21126 (NanoSpace) and CA21106 (CosmicWISPers), 2023–present.
- Tutor Buddy for international students, University of Padua.

References

Prof. Eugenio Bottacini
Thesis supervisor
Department of Physics and Astronomy
University of Padua, Italy

Dr. Andrea Tramacere
Thesis co-supervisor
Department of Astronomy
University of Geneva, Switzerland

Contact details available on request.

Other Interests

- **Sociophysics** (independent project): modelling political opinion dynamics as motion on a potential landscape, with the potential estimated from longitudinal survey data (ANES).
- **Photography:** event photography (weddings, graduations, portraits) since 2024, and travel photography around northern Italy.